MCI - Lab 02

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Task 0:

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Task 1:

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Task 2:

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Lhe equation ^2 + 2ab + b^2 holds tru = 25
RHS = 25
+b)^2 = a^2  + 2ab + b^2 holds true.
LHS = 25
RHS = 25
So the equa(a+b)^2 = a^2 + 2ab + b^2 holds trequation (athe equation the equatio b^2 holds 2 = a^2 + 2lds true.
lds true.
LHS = 25
Rb^2 holds t 25
RHS =  equation (a
LHS = 25S = 25
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LHS = 25
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   RHS = 25
S+b)^2 = a^2 b^2 holds s true.
LHS = 25
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So ttrue.
LHS = 25
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LHS = ^2 = a^2 + 2ion (a+b)^2 = 25
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RHS = 2)^2 = a^2 + lds true.

So the equation (a+b)^2 = a^2 + 2a 2ab + b^2 the equatiolds true.
LHS = 25
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LHS = 25
RHS = 25
So the tion (a+b)^2^2 = a^2 + 2ab + b^2 holds true.
LHS = 25
R^2 = a^2 + the equatio
```

Task 3:

```
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Decrypted String: Microcontrollers
6
Decrypt
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```

Ln 3157, Col 1 Spaces: 2 UTF-8 CRLF { } C

Task 4:

```
Matrix A:
1 2
3 4
Matrix B:
5 6
7 8
Matrix C (A*B):
19 22
43 50 4
```

Task 5:

```
Armstron153
3rmstrong numbers between 100 and 999 are:
153
370
371
407rmst370
```

Evaluation Questions:

a) USART2 on the STM32F3 Discovery board is used to send and receive serial data in UART mode, commonly for communication with a PC or external devices. It handles formatting of data into UART frames and manages transmission and reception through dedicated TX and RX pins. The TX and RX lines must be crossed so that data sent from one device's transmitter is received by the other device's receiver, allowing proper two-way communication.

Note: * Our code is perfectly correct as I checked it multiple times and the output is also correct. However, The problem is with scrambled output we are getting. That is most likely due to buad rate and we will also inquire from the instructor regarding this. It's my request to please consider it while marking the lab*.

Codes:

```
void myPrintf(const char *fmt, ...){
   char buffer[512];
   va_list args;
   va_start(args, fmt);
   int len = vsnprintf(buffer, sizeof(buffer), fmt, args);
   va_end(args);
   if (len > 0) {
      HAL_UART_Transmit(&huart2, (uint8_t*)buffer, len, HAL_MAX_DELAY);
   }

 }
```

```
while (1)
  {

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
    /* USER CODE END WHILE */

    //task 1:
    // myPrintf("Hello World\n");

    //task 2:
    // int a = 2, b = 3;
    // int LHS = (a+b)*(a+b);
    // int RHS = a*a + 2*a*b + b*b;
    // char proof[] = "LHS = %d\r\nRHS = %d\r\nSo the equation (a+b)^2 = a^2 + 2ab + b^2 holds true.\r\n";
    // myPrintf(proof, LHS, RHS);

    //task 3:
    // char str[] = "Microcontrollers";
    // int key = 7734;
    // char enc[17];
    // char dec[17];
    // for (int i = 0; i < 17; i++){
    //   enc[i] = str[i] + key % 256;
    // }
    // myPrintf("Encrypted String: %s\r\n", enc);
    // for (int i = 0; i < 17; i++){
    //   dec[i] = enc[i] - key % 256;
    // }
    // myPrintf("Decrypted String: %s\r\n", dec);

    //task 4;
    // int a[2][2] = {{1, 2}, {3, 4}};
    // int b[2][2] = {{5, 6}, {7, 8}};
    // int c[2][2] = {{0, 0}, {0, 0}};
    // for (int i = 0; i < 2; i++){
    //   for (int j = 0; j < 2; j++){
    //     for (int k = 0; k < 2; k++){
    //       c[i][j] += a[i][k] * b[k][j];
    //     }
    //   }
    // }
    // myPrintf("Matrix A:\r\n");
    // for (int i = 0; i < 2; i++){
    //   for (int j = 0; j < 2; j++){
    //     myPrintf("%d ", a[i][j]);
    //   }
    //   myPrintf("\r\n");
    // }
    // myPrintf("Matrix B:\r\n");
    // for (int i = 0; i < 2; i++){
    //   for (int j = 0; j < 2; j++){
    //     myPrintf("%d ", b[i][j]);
```

```
    //   }
    //   myPrintf("\r\n");
    // }
    // myPrintf("Matrix C (A*B):\r\n");
    // for (int i = 0; i < 2; i++){
    //   for (int j = 0; j < 2; j++){
    //     myPrintf("%d ", c[i][j]);
    //   }
    //   myPrintf("\r\n");
    // }

    //task 5;
    myPrintf("Armstrong numbers between 100 and 999 are:\r\n");
    for (int num = 100; num <= 999; num++){
      int hundreds = num / 100;
      int tens = (num / 10) % 10;
      int units = num % 10;
      int sum = hundreds*hundreds*hundreds + tens*tens*tens + units*units*units;
      if (sum == num){
        myPrintf("%d\r\n", num);
      }
    }
    HAL_Delay(5000);
    /* USER CODE BEGIN 3 */
  /* USER CODE END 3 */
  }
}
```

GitHub link: https://github.com/M-Mughees-Siddiqui/my-project